



FRONTEND DEVELOPMENT - II

CSC 1051: FULL-STACK FUNDAMENTALS LABORATORY

N.H.Wanigasingha (MPhil (Reading), BSc (Hons))

Department of Computer Science

Faculty of Applied Sciences

University of Sri Jayewardenepura

INTRODUCTION TO CSS

- CSS stands for **C**ascading **S**tyle **S**heets.
- It is used to style HTML elements.
- CSS describes how elements are displayed on the screen, paper, or other media.

WHY USE CSS?

1. Separate content from presentation.
2. Make websites visually appealing.
3. Easier to maintain and update.
4. Provide consistent styling across pages.
5. Optimize websites for different devices and screen sizes.

TYPES OF CSS

Inline

```
<tag style="css" />
```

Internal

```
<style>css</style>
```

External

```
<link href="style.css"/>
```

1. Inline CSS: Applied directly to an HTML element.
2. Internal CSS: Defined within a <style> tag in the <head> section.
3. External CSS: Linked as a separate .css file.

INLINE CSS

Property Value

```
<html style="background: blue">
</html>
```

all tag

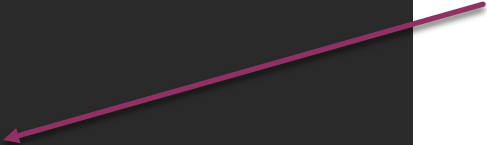
```
<html style="background: blue">
</html>
```

name value

INTERNAL CSS

```
<html>
<head>
  <style>
    html {
      background: red;
    }
  </style>
</head>
</html>
```

CSS
Selector



EXTERNAL CSS

```
index.html x
<html>
  <head>
    <link
      rel="stylesheet"
      href="./styles.css"
    />
  </head>
</html>

styles.css x
html {
  background: green;
}
```

Linking Element

Relationship

Location

Value

Property

COMPARISON

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="UTF-8">
6   <title>Inline</title>
7 </head>
8
9 <body>
10  <h1 style="color: blue;">Style Me in Blue!</h1>
11 </body>
12
13 </html>
```

Inline

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="UTF-8">
6   <title>Internal</title>
7   <style>
8     h1 {
9       color: red;
10    }
11  </style>
12 </head>
13
14 <body>
15   <h1>Style Me in Red!</h1>
16 </body>
17
18 </html>
```

Internal

COMPARISON

```
1. Adding CSS > Solution > external.html > html
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <title>External</title>
7      <link rel="stylesheet" href="./style.css" />
8  </head>
9
10 <body>
11     <h1>Style Me in Green</h1>
12 </body>
13
14 </html>
```

.html File

```
1  h1 {
2      color: green;
3  }
4
```

.css File

COMPARISON

- Inline: one element
- Internal: one webpage
- External: one website (multi page website)

CSS SELECTORS



A handwritten diagram on a light gray background. At the top, the words "CSS Selector" are written in purple cursive. A purple line starts from the "C" and curves down to a purple circle that encloses the text "h1{". Below this, the text "color: blue" is written in blue and red, and a closing curly brace "}" is written in orange at the bottom.

```
h1{  
  color: blue  
}
```

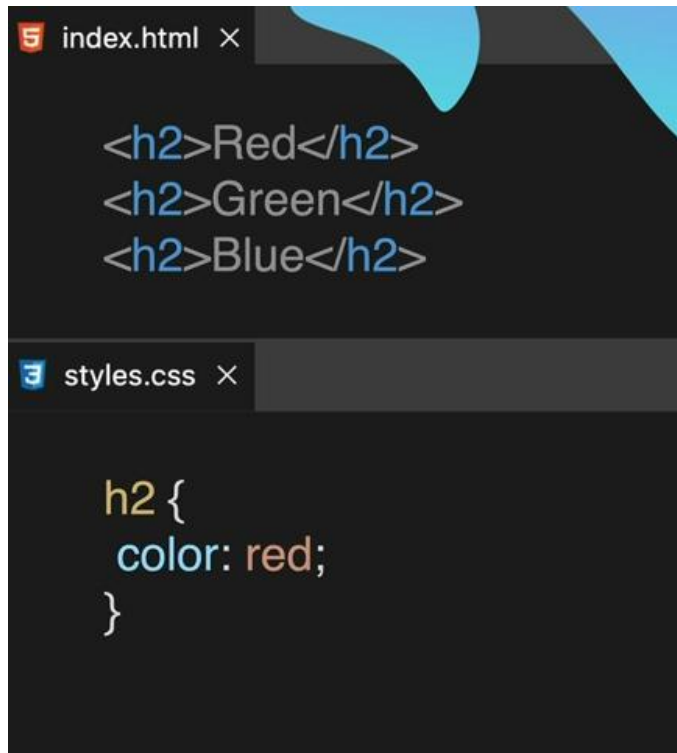
Selects the part of the HTML where the CSS rules are to be applied.

ELEMENT SELECTOR

```
h1 {  
  color: blue  
}
```

Selects all elements with the specified tag.

ELEMENT SELECTOR



The screenshot shows a code editor with two files open. The top file, 'index.html', contains three `<h2>` tags with the text 'Red', 'Green', and 'Blue'. The bottom file, 'styles.css', contains a CSS rule for `h2` with the property `color: red;`.

```
index.html ×  
<h2>Red</h2>  
<h2>Green</h2>  
<h2>Blue</h2>  
  
styles.css ×  
  
h2 {  
  color: red;  
}
```

Red

Green

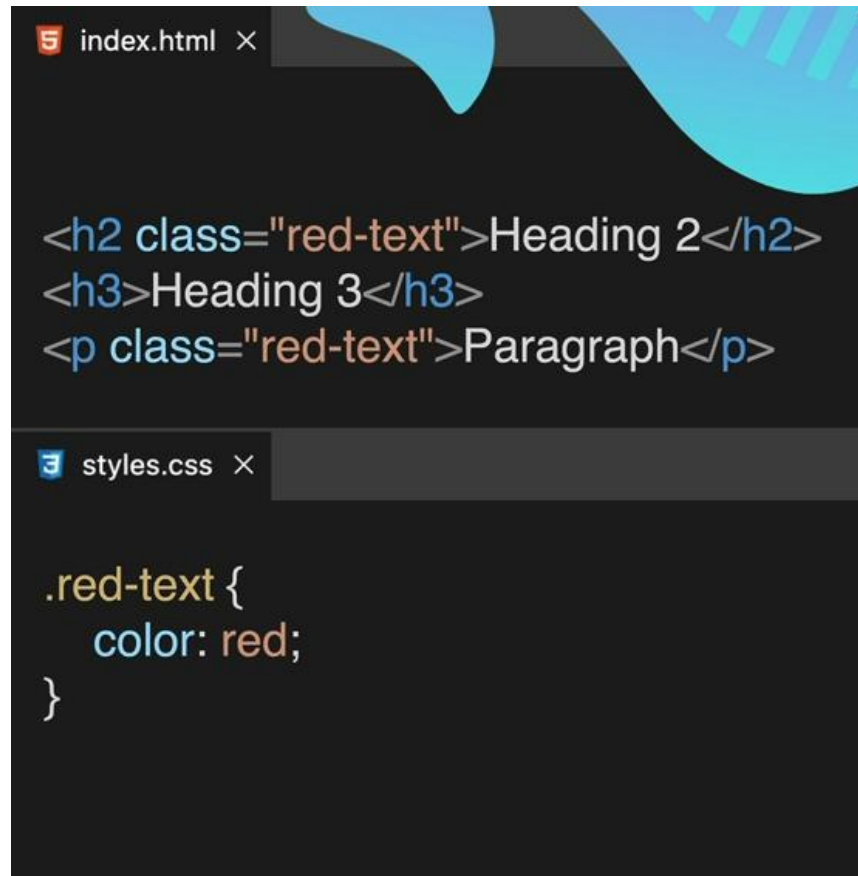
Blue

Selects all elements with the specified tag.

CLASS SELECTOR

```
.red-heading {  
  color: red  
}
```

CLASS SELECTOR



The image shows a code editor with two tabs: 'index.html' and 'styles.css'. The 'index.html' tab is active, displaying the following HTML code:

```
<h2 class="red-text">Heading 2</h2>
<h3>Heading 3</h3>
<p class="red-text">Paragraph</p>
```

The 'styles.css' tab is also visible, displaying the following CSS code:

```
.red-text {
  color: red;
}
```

The code is color-coded: HTML tags are blue, attributes and values are orange, and CSS selectors and values are yellow. The background of the code editor is dark, and there are some blue and green abstract shapes in the top right corner.

ID SELECTOR

A screenshot of a code editor with two tabs: 'index.html' and 'styles.css'. The 'index.html' tab is active, showing three lines of HTML code: `<h2 id="main">Red</h2>`, `<h2>Green</h2>`, and `<h2>Blue</h2>`. The 'styles.css' tab is also visible, showing a CSS rule: `#main { color: red; }`.

```
index.html ×  
<h2 id="main">Red</h2>  
<h2>Green</h2>  
<h2>Blue</h2>  
  
styles.css ×  
#main {  
  color: red;  
}
```

Red

Green

Blue

CLASS VS ID SELECTOR

- ID should be kept unique for an element. Styling is applied only for that element.
- Class can be applied for many elements.

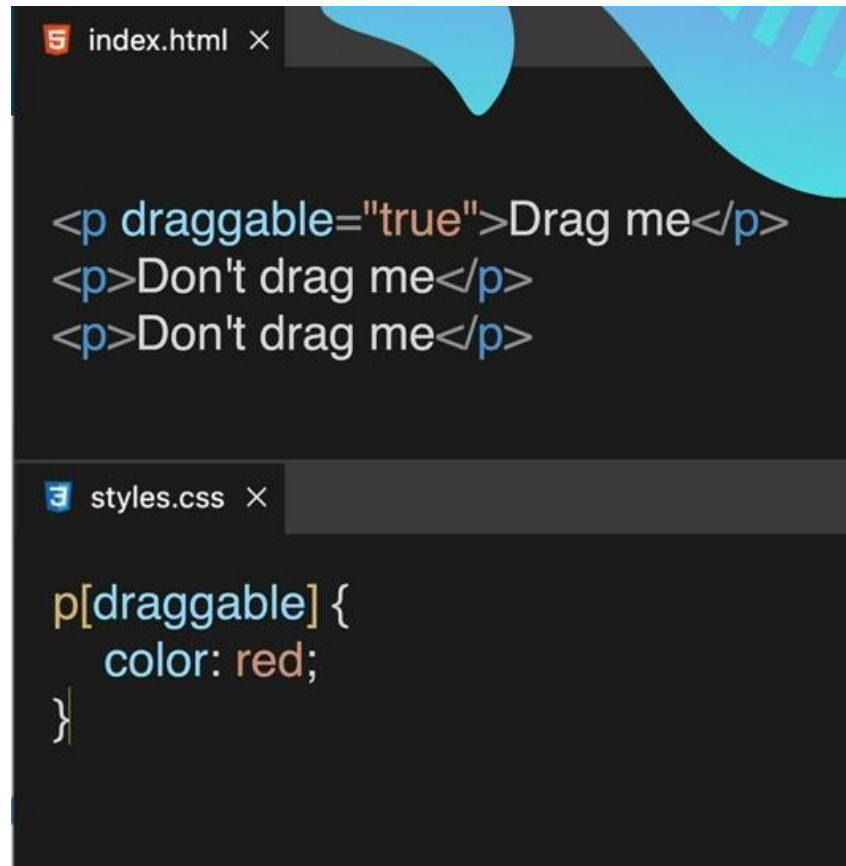
ATTRIBUTE SELECTOR



The diagram shows a CSS attribute selector rule: `p[draggable]{ color: red }`. Handwritten purple annotations include an arrow pointing from the text "html element" to the `p` selector, and another arrow pointing from the text "attribute" to the `draggable` attribute in the selector.

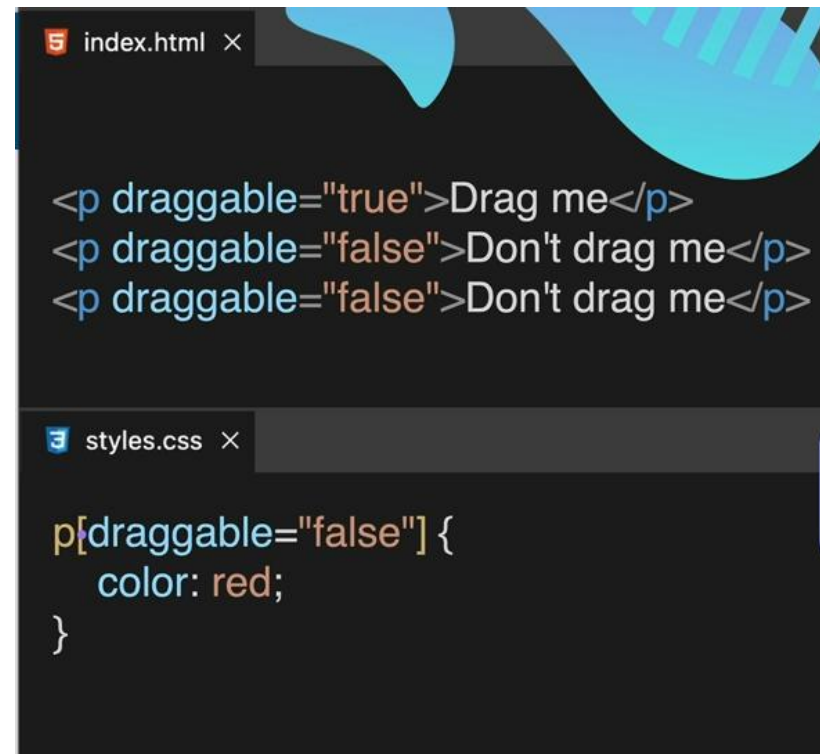
```
p[draggable]{  
  color: red;  
}
```

ATTRIBUTE SELECTOR



A screenshot of a code editor with two tabs: 'index.html' and 'styles.css'. The 'index.html' tab is active, showing three lines of HTML code: `<p draggable="true">Drag me</p>`, `<p>Don't drag me</p>`, and `<p>Don't drag me</p>`. The 'styles.css' tab is also visible, showing a CSS rule: `p[draggable] { color: red; }`.

```
index.html ×  
  
<p draggable="true">Drag me</p>  
<p>Don't drag me</p>  
<p>Don't drag me</p>  
  
styles.css ×  
  
p[draggable] {  
  color: red;  
}
```



A screenshot of a code editor with two tabs: 'index.html' and 'styles.css'. The 'index.html' tab is active, showing three lines of HTML code: `<p draggable="true">Drag me</p>`, `<p draggable="false">Don't drag me</p>`, and `<p draggable="false">Don't drag me</p>`. The 'styles.css' tab is also visible, showing a CSS rule: `p[draggable="false"] { color: red; }`.

```
index.html ×  
  
<p draggable="true">Drag me</p>  
<p draggable="false">Don't drag me</p>  
<p draggable="false">Don't drag me</p>  
  
styles.css ×  
  
p[draggable="false"] {  
  color: red;  
}
```

BOX MODEL

```
height: 600px;  
width: 600px;
```

BOX MODEL - BORDER



A diagram illustrating the syntax of the CSS border property. The text `border: 10px_solid_black;` is shown. Above the text, three handwritten labels in purple ink are connected to the corresponding parts of the code by lines: "thickness" points to "10px", "style" points to "solid", and "Color" points to "black".

```
border: 10px_solid_black;
```

BOX MODEL - BORDER

- border: 30px solid black;
 - border-top: 0px;

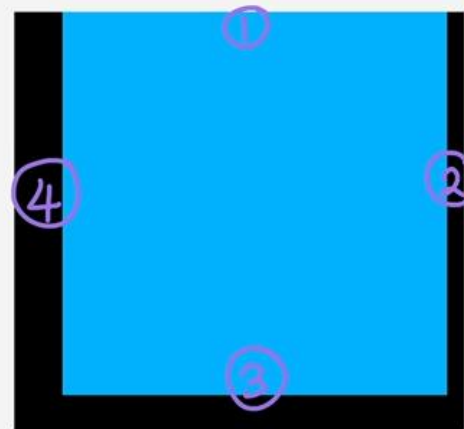


BOX MODEL - BORDER

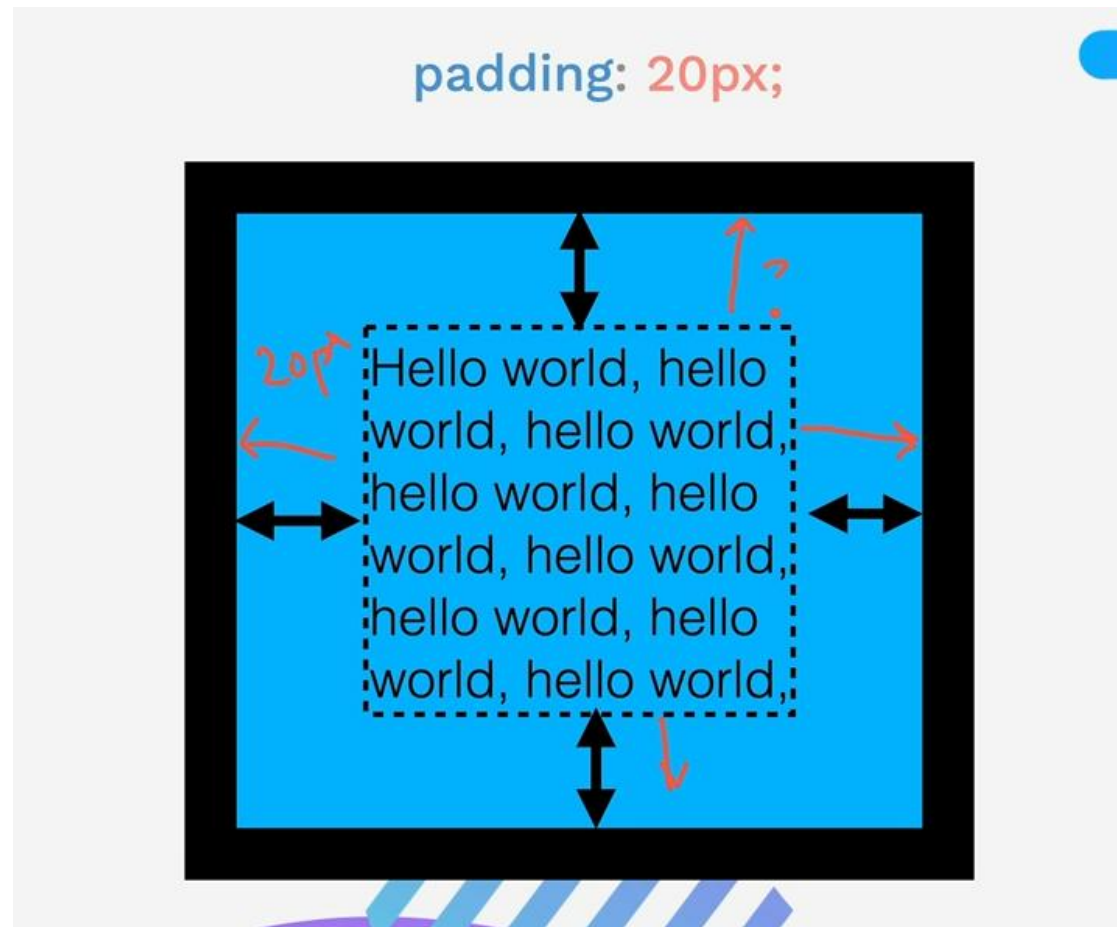
- border: 30px solid black;
 - border-top: 0px;



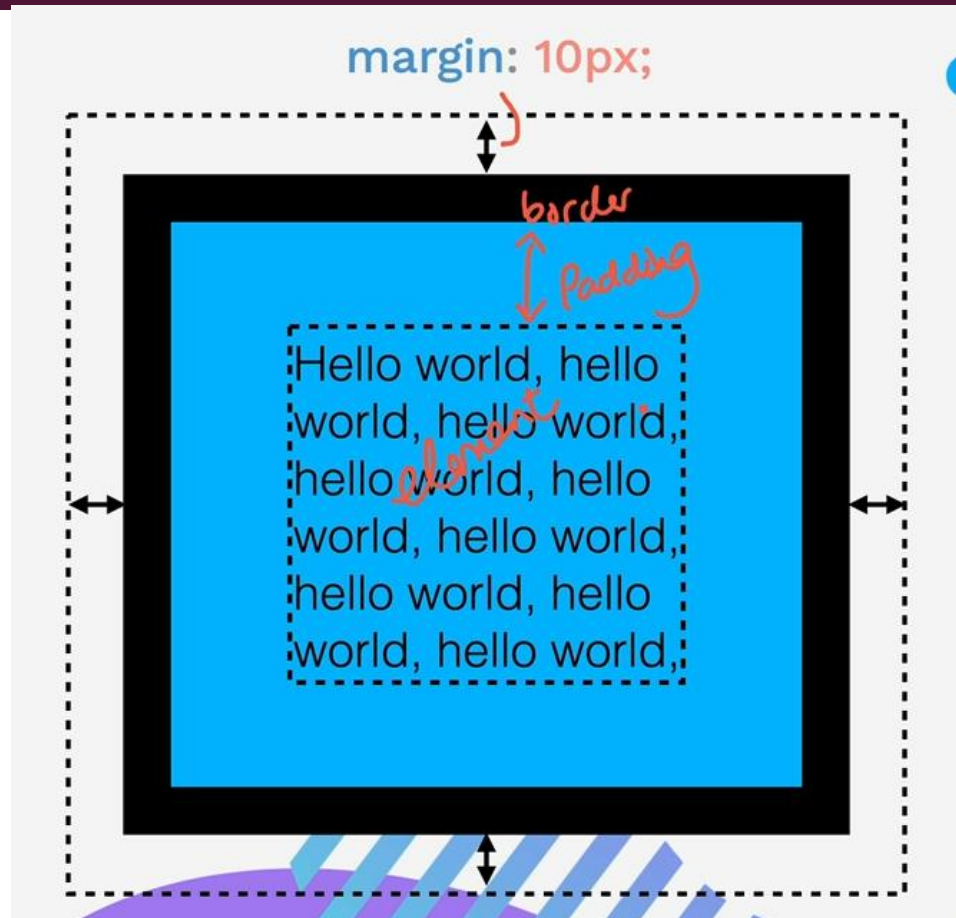
border: 30px solid black;
border-width: 0px 10px 20px 30px;



BOX MODEL - PADDING



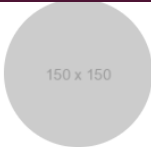
BOX MODEL - MARGIN



CSS DOCUMENTATION

<https://developer.mozilla.org/en-US/docs/Web/CSS>

CSS LAB EXERCISE: APPLY CSS TO PERSONAL PORTFOLIO WEBPAGE



Name

Web Developer | Designer | Problem Solver

About Me

Hello! I'm John, a passionate web developer with experience in creating dynamic and responsive websites. I love coding and solving problems creatively.

Skills

- HTML & CSS
- JavaScript
- React.js
- Node.js
- Python

Projects

Project Name	Description	Technology Used	Preview
Portfolio Website	A personal website to showcase my skills and projects.	HTML, CSS	
E-commerce App	An online shopping platform with cart functionality.	React, Node.js	
Blog Platform	A platform for users to share their stories and posts.	Python, Flask	

Contact Me

Name:

Email:

Message:

Connect with me:

[GitHub](#) | [LinkedIn](#)

CSS LAB EXERCISE: APPLY CSS TO PERSONAL PORTFOLIO WEBPAGE

150 x 150

Name

Web Developer | Designer | Problem Solver

About Me

Hello! I'm John, a passionate web developer with experience in creating dynamic and responsive websites. I love coding and solving problems creatively.

Skills

- HTML & CSS
- JavaScript
- React.js
- Node.js
- Python

Projects

Project Name	Description	Technology Used	Preview
Portfolio Website	A personal website to showcase my skills and projects.	HTML, CSS	100 x 100
E-commerce App	An online shopping platform with cart functionality.	React, Node.js	100 x 100
Blog Platform	A platform for users to share their stories and posts.	Python, Flask	100 x 100

CSS LAB EXERCISE: APPLY CSS TO PERSONAL PORTFOLIO WEBPAGE

Contact Me

Name:

Email:

Message:

Submit

Connect with me:

[GitHub](#) | [LinkedIn](#)